RVS TECHNICAL CAMPUS – COIMBATORE

Department of Computer Science and Engineering

BATCH – 2022-2026 **Semester VIII**

REVIEW - III

ML-Based College Recommendation System for TNEA Aspirants

Presented By

- Naveen Rajan N - 712922104035
- Sabarimani D - 712922104043
- Sanjay Y - 712922104046
- Satheesh - 712922104305

Guided by,

Mrs. R.Vimala

Assistant Professor

Department of CSE

RVS - TCC

Introduction

- Tamil Nadu Engineering Admissions (TNEA) counseling involves complex decision-making based on cutoff marks.
- Students often face difficulty in:
  - Analyzing past cutoff trends
  - Selecting suitable colleges and branches
- Existing data sources are static and not personalized.
- This project introduces a Machine Learning–powered web platform to assist students with data-driven insights and recommendations.

Problem Statement

Students currently lack an intelligent system to guide them during the TNEA counseling process. Manual analysis of cutoff PDFs and websites is time-consuming and error-prone. There is no personalization based on student cutoff, community, or preferences. Hence, there is a need for a reliable, explainable, and scalable ML-based recommendation system.

Objectives

- Analyze historical TNEA cutoff data using Machine Learning techniques.
- Predict cutoff trends and reasonable ranges for upcoming years.
- Recommend suitable college–branch options based on student profile.
- Rank recommendations by likelihood of admission.
- Provide an interactive and user-friendly web interface.

Literature Survey

- Previous systems focused mainly on static cutoff display.
- Regression models have been used for academic cutoff prediction with limited accuracy.
- Recommendation systems are widely adopted for personalized decision support.
- Recent research emphasizes:
  - Hybrid ML approaches
  - Explainable and data-driven recommendations
- Limitations identified: lack of personalization and real-world applicability.

Existing System

- Students refer to:
- Government PDFs
- Informational websites
- No Machine Learning involvement.
- No prediction or ranking mechanism.
- No personalized guidance based on student cutoff and community.
- Limited visualization and decision support.

Proposed System

- An ML-powered college recommendation platform.
- Uses historical cutoff data to:
  - Predict trends
  - Estimate admission probability
  - Rank college–branch options
- Integrated web application with:
  - Dashboards
  - Chatbot support
- Designed for accuracy, explainability, and scalability.

Machine Learning Pipeline

- 1. Trend Forecasting (Regression)
  - Model: LightGBM (Gradient Boosted Decision Trees)
  - Objective: Predict next-year cutoff
  - Captures non-linear feature interactions
  - Validation MAE: 0.76 marks
- 2. Admission Probability (Classification)
  - Model: Logistic Regression
  - Input: Score gap vs predicted cutoff
  - Output: Calibrated probability (0–1)
  - Uses Sigmoid function for probabilistic mapping
- 3. Recommendation Layer
  - Safe: Probability > 0.75
  - Moderate: 0.40–0.75
  - Ambitious: < 0.40

Machine Learning Pipeline for Admission Prediction

Data Modeling

Creating a structured dataset for analysis

Feature Engineering

Developing relevant features for prediction

Trend Forecasting

Predicting next-year cutoff using LightGBM

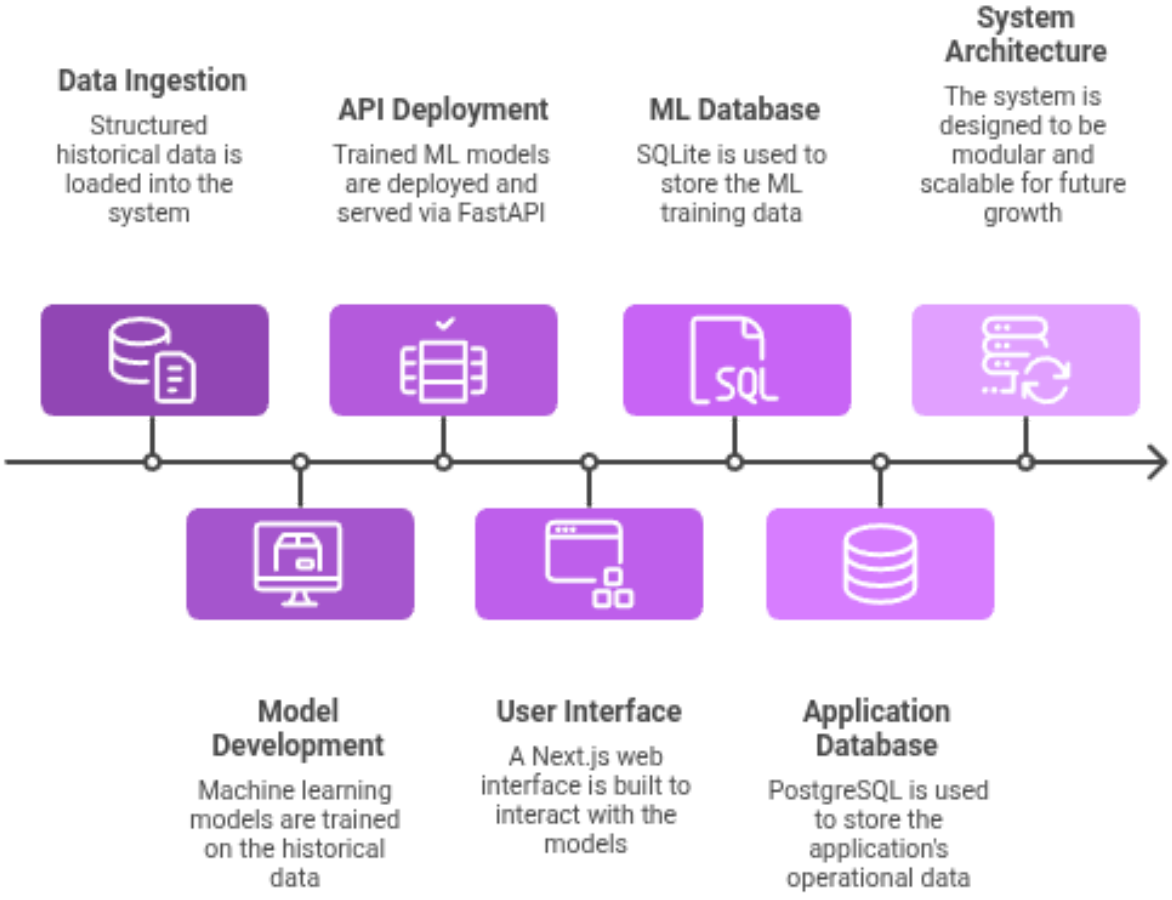
Admission Probability

Calculating admission probability using Logistic Regression

Recommendation Layer

Categorizing recommendations based on probability

System Architecture



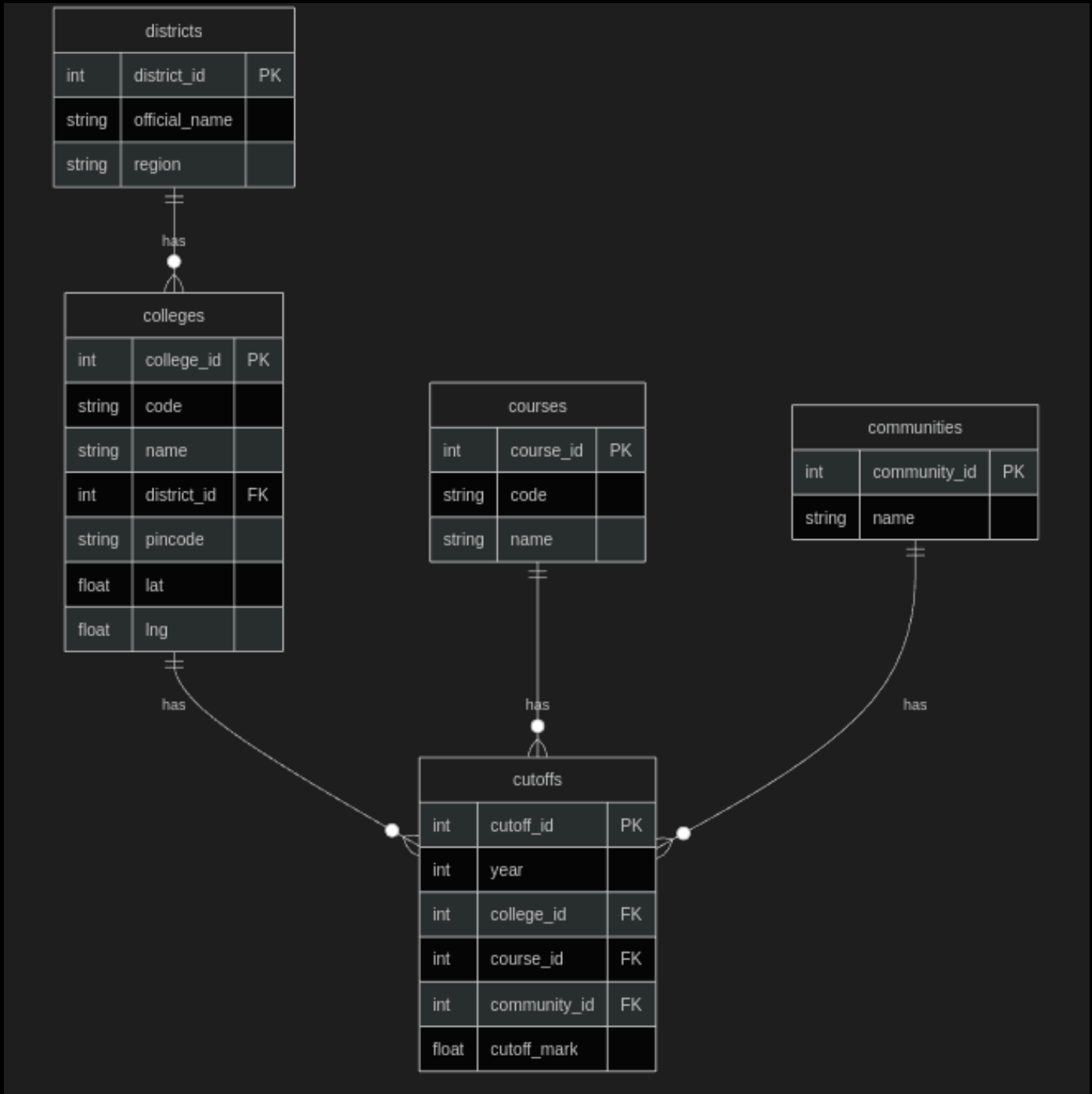
Module Description

- Module 1: Data Processing Module
  - Cleans and normalizes cutoff data.
  - Organizes data by college, branch, year, and community.
- Module 2: ML Prediction & Recommendation Module
  - Predicts cutoff trends and computes admission probability.
  - Ranks college–branch combinations based on likelihood.
- Module 3: Web Application & Visualization Module
  - Handles user interaction and processes student inputs.
  - Displays recommendations with interactive visualizations.
- Module 4: Intelligent Chat Assistant Module
  - Supports natural language queries on cutoff data.
  - Provides dynamic, personalized insights conversationally.

Module 1 Result

- Successfully cleaned and structured historical cutoff dataset.
- Improved data consistency across years.
- Enabled efficient querying for ML models.

DB Schema Structure



Module 2 Result

- Generated cutoff trend predictions within acceptable error margins.
- Produced ranked college–branch recommendations.
- Provided admission probability scores to support decision-making.

```
nr@fedora:~/projects/clg$ python3 recommend.py 185.0 BC
Analyzing 2280 options for Score: 185.0, Community: BC...

Summary:
  SAFE ✅: 2157 options
  MODERATE ⚠️: 5 options
  AMBITIOUS 🚀: 118 options

--- SAFE ✅ ---
                                                                                                                college_name                                                  course_name  pred_2025_cutoff  admission_prob
                                                Latha Mathavan Engineering College, Latha Mathavan Nagar, Kidaripatti, Melur Taluk, Madurai-625301.                   Computer Science And Engineering         86.642648             1.0
Theni Kammavar Sangam College of Technology, Theni Main road, Koduvilarpatti (PO), Veerapandi(Via), Theni District-625534.                                         Information Technology         86.238000             1.0
Theni Kammavar Sangam College of Technology, Theni Main road, Koduvilarpatti (PO), Veerapandi(Via), Theni District-625534. Computer Science And Engineering (Ai And Machine Learning)         91.029799             1.0
Theni Kammavar Sangam College of Technology, Theni Main road, Koduvilarpatti (PO), Veerapandi(Via), Theni District-625534.                 Electronics And Communication Engineering        103.121672             1.0
                                                SBM College of Engineering and Technology, SBM Nagar, Thamaraipadi(PO), Trichy Road, Dindigul-624005.                   Computer Science And Engineering         81.049723             1.0
Syed Ammal Engineering College (Autonomous), Dr. E.M.Abdullah Campus, Lanthai, Achunthanvayal Post, Ramanathapuram-623502.                                         Bio Medical Engineering        106.981740             1.0
Syed Ammal Engineering College (Autonomous), Dr. E.M.Abdullah Campus, Lanthai, Achunthanvayal Post, Ramanathapuram-623502.                             Computer Science And Bussiness System        130.272317             1.0
Syed Ammal Engineering College (Autonomous), Dr. E.M.Abdullah Campus, Lanthai, Achunthanvayal Post, Ramanathapuram-623502.                          Artificial Intelligence And Data Science        136.445650             1.0
Syed Ammal Engineering College (Autonomous), Dr. E.M.Abdullah Campus, Lanthai, Achunthanvayal Post, Ramanathapuram-623502.                                         Information Technology        135.951636             1.0
Syed Ammal Engineering College (Autonomous), Dr. E.M.Abdullah Campus, Lanthai, Achunthanvayal Post, Ramanathapuram-623502.                                         Mechanical Engineering         89.267311             1.0

--- MODERATE ⚠️---
                                                                                                                college_name                                                  course_name  pred_2025_cutoff  admission_prob
                                   Rajalakshmi Engineering College  (Autonomous), Rajalakshmi Nagar, Thandalam, Kanchipuram, Chennai-602105.                 Electronics And Communication Engineering        184.937872        0.718725
KPR Institute of Engineering and Technology (Autonomous), SF:204/2 & 204/4, Kollupalayam Village, Arasur Panchayat, Coimbatore-641407.                                 Computer Science And Engineering        184.957532        0.666898
KPR Institute of Engineering and Technology (Autonomous), SF:204/2 & 204/4, Kollupalayam Village, Arasur Panchayat, Coimbatore-641407. Computer Science And Engineering (Ai And Machine Learning)        184.949731        0.665738
                                  Chennai Institute of Technology (Autonomous) Puduper Village  Nandambakkam Post  Kundrathur  Chennai 600069                                Mechatronics Engineering        184.971097        0.599115
                                           Coimbatore Institute of Technology (Autonomous)  Civil Aerodrome Post  Coimbatore District 641014                                    Chemical  Engineering        184.971035        0.597958

--- AMBITIOUS 🚀 ---
                                                                                                                college_name                                                  course_name  pred_2025_cutoff  admission_prob
                                   Rajalakshmi Engineering College  (Autonomous), Rajalakshmi Nagar, Thandalam, Kanchipuram, Chennai-602105.                         Information Technology        185.055388        0.306685
                                Sri Krishna College of Engineering and Technology (Autonomous), Sugunapuram, Kuniamuthur Post, Coimbatore-641008.                       Computer Science And Design        185.088199        0.205112
                                Sri Venkateswara College of Engineering (Autonomous), Post Bag No.3, Pennalur, Sriperumbudur, Kanchipuram-602105.  Artificial Intelligence And Data Science        185.110342        0.161672
                                           PSG Institute of Technology and Applied Research, Avinashi Road,Neelambur, Sulur,Coimbatore -641062.                         Mechanical Engineering        185.168095        0.079191
Prince Shri Venkateshwaraa Padmavathy Engineering College (Autonomous), Medavakkam - Mambakkam Road, Ponmar, Kanchipuram, Chennai-600127.                         Information Technology        185.192047        0.066844
                                   Rajalakshmi Engineering College  (Autonomous), Rajalakshmi Nagar, Thandalam, Kanchipuram, Chennai-602105.  Artificial Intelligence And Data Science        185.189667        0.057015
                                                   Government College of Engineering (Autonomous)  Karuppur  Salem District 636011    Electrical And Electronics Engineering        185.311095        0.010872
                                         Thiagarajar College of Engineering (Autonomous) Tirupparankundram  Madurai District 625015                         Mechanical Engineering        185.585720        0.000195
                                Sri Venkateswara College of Engineering (Autonomous), Post Bag No.3, Pennalur, Sriperumbudur, Kanchipuram-602105. Electronics And Communication Engineering        185.635281        0.000099
                                         Thiagarajar College of Engineering (Autonomous) Tirupparankundram  Madurai District 625015                               Mechatronics (Ss)        185.871116        0.000003
```

Module 3 Result

- User-friendly web interface developed for seamless interaction with cutoff data.
- Efficient processing of student inputs (cutoff, community, preferences).
- ML-based recommendations (safe, moderate, ambitious) displayed clearly.
- Interactive visualizations (trend graphs, regional distribution, branch analysis).

TNEA Cutoff Analytics

Statistical analysis of Tamil Nadu Engineering Admissions historical data (2020-2024).

Cutoff Mark (Aggregate)

185

Community Group

Backward Class

Analyze Probabilities

SAFE

Latha Mathavan Engineering College, Latha Mathavan Nagar, Kidaripatti, Melur Taluk, Madurai-625301.
Computer Science And Engineering
Projected: 86.64 — 100.0% Prob.

Theni Kammavar Sangam College of Technology, Theni Main road, Koduvilarpatti (PO), Veerapandi(Via), Theni District-625534.
Information Technology
Projected: 86.24 — 100.0% Prob.

University Departments of Anna University Chennai - CEG Campus Sardar Patel Road Guindy Chennai 600 025
Printing & Packing Technology
Projected: 175.89 — 100.0% Prob.

University Departments of Anna University Chennai - CEG Campus Sardar

MODERATE

Rajalakshmi Engineering College (Autonomous), Rajalakshmi Nagar, Thandalam, Kanchipuram, Chennai-602105.
Electronics And Communication Engineering
Projected: 184.94 — 71.9% Prob.

KPR Institute of Engineering and Technology (Autonomous), SF:204/2 & 204/4, Kollupalayam Village, Arasur Panchayat, Coimbatore-641407.
Computer Science And Engineering
Projected: 184.96 — 66.7% Prob.

KPR Institute of Engineering and Technology (Autonomous), SF:204/2 & 204/4, Kollupalayam Village, Arasur Panchayat, Coimbatore-641407.
Computer Science And Engineering (Ai And Machine Learning)
Projected: 184.95 — 66.6% Prob.

AMBITIOUS

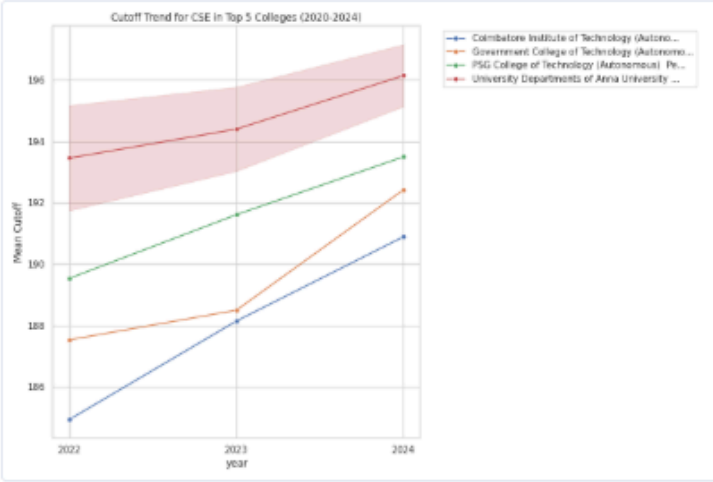
Rajalakshmi Engineering College (Autonomous), Rajalakshmi Nagar, Thandalam, Kanchipuram, Chennai-602105.
Information Technology
Projected: 185.06 — 30.7% Prob.

Sri Krishna College of Engineering and Technology (Autonomous), Sugunapuram, Kuniamuthur Post, Coimbatore-641008.
Computer Science And Design
Projected: 185.09 — 20.5% Prob.

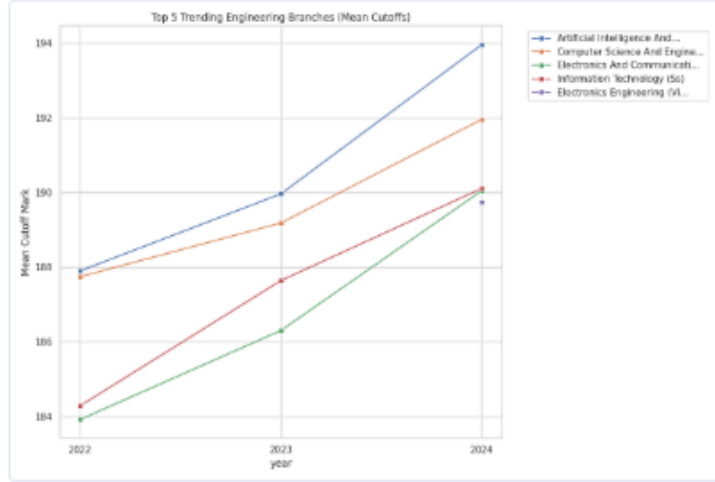
Sri Venkateswara College of Engineering (Autonomous), Post Bag No.3, Pennalur, Sriperumbudur, Kanchipuram-602105.
Artificial Intelligence And Data Science
Projected: 185.11 — 16.2% Prob.

PSG Institute of Technology and Applied

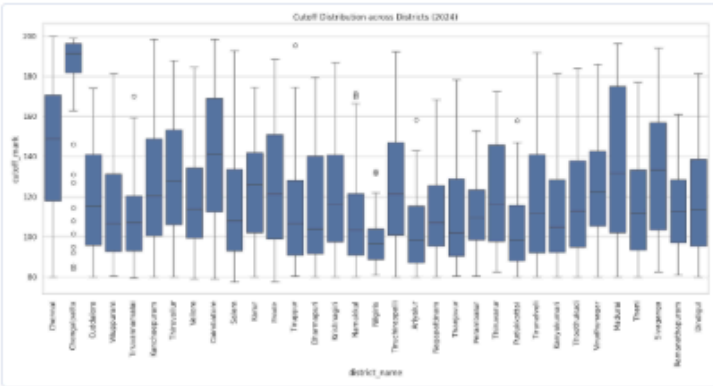
Trend Visualizations



Department Specific Trends — Historical year-over-year performance for key engineering branches.



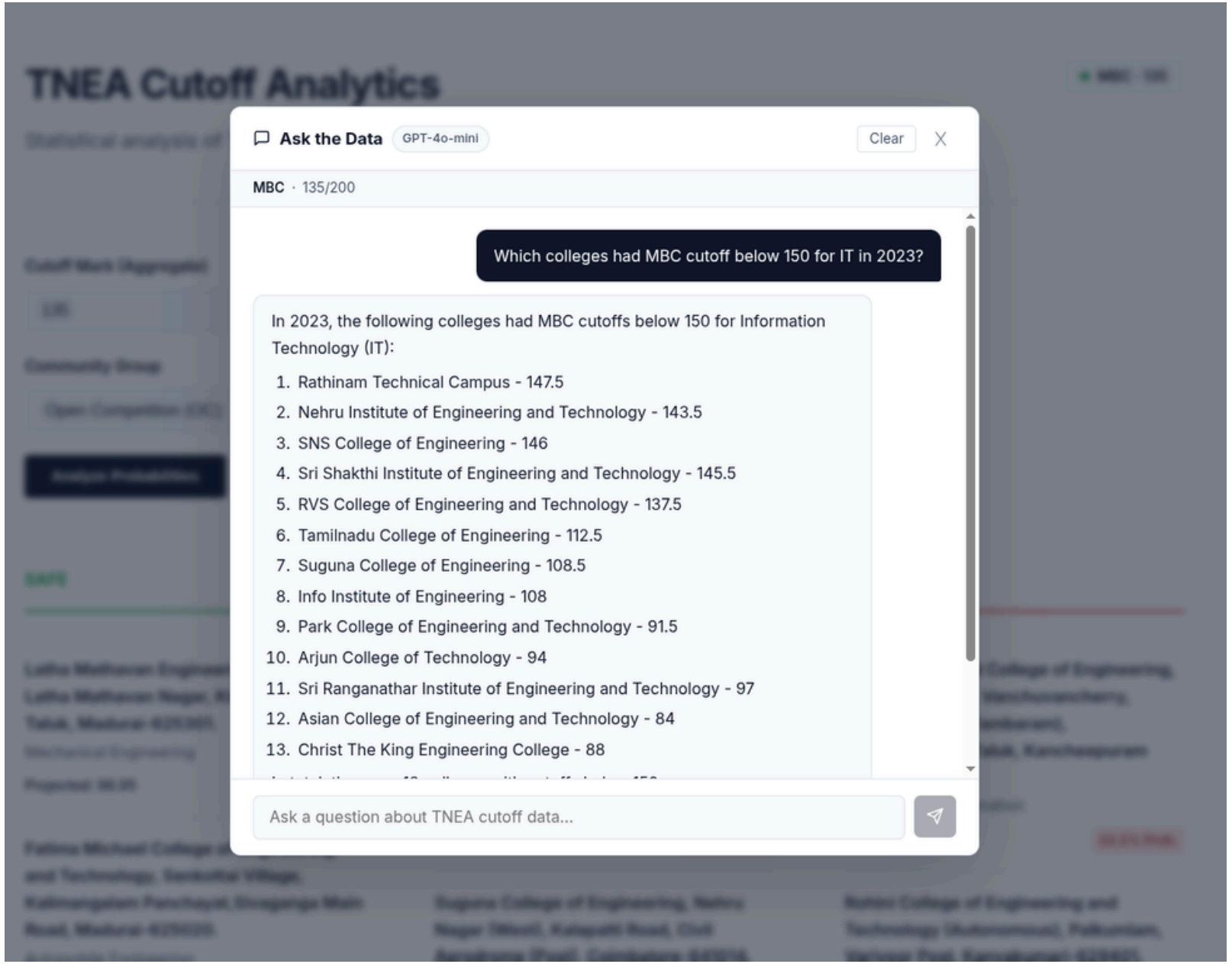
Sector Demand Variance — Analysis of shifting cutoff thresholds across major engineering disciplines.



Regional Distribution — Geographic distribution of institutional performance across TN districts.

Module 4 Result

- Natural language query system for accessing cutoff insights .
- Dynamic retrieval of data based on user questions and filters.
- Context-aware responses integrating historical data and ML predictions.
- Supports follow-up queries for refined and personalized results.
- Enhances user experience through conversational decision support.


TNEA Cutoff Analytics
Ask the Data
GPT-4o-mini
Clear
MBC · 135/200
Which colleges had MBC cutoff below 150 for IT in 2023?
In 2023, the following colleges had MBC cutoffs below 150 for Information Technology (IT):
1. Rathinam Technical Campus - 147.5
2. Nehru Institute of Engineering and Technology - 143.5
3. SNS College of Engineering - 146
4. Sri Shakthi Institute of Engineering and Technology - 145.5
5. RVS College of Engineering and Technology - 137.5
6. Tamilnadu College of Engineering - 112.5
7. Suguna College of Engineering - 108.5
8. Info Institute of Engineering - 108
9. Park College of Engineering and Technology - 91.5
10. Arjun College of Technology - 94
11. Sri Ranganathar Institute of Engineering and Technology - 97
12. Asian College of Engineering and Technology - 84
13. Christ The King Engineering College - 88
Ask a question about TNEA cutoff data...

Conclusion

The project presents a Machine Learning–driven decision support system for TNEA aspirants. By combining predictive modeling, recommendation techniques, and explainable analytics, it helps students make informed choices during counseling. The system is scalable, extensible, and suitable for real-world deployment.